10. Quantum Black Holes

Black holes populate our universe in great number. They are regions in which space is so curved as to collapse in on itself, and where time comes to a standstill. As mentioned, they form, for instance, when a star has burned up all of the available hydrogen and collapses.

Frequently, the collapsed star formed part of a pair of neighbouring stars and, in this case, the black hole and the surviving counterpart circle one around the other; the black hole sucks matter from the other star continuously (as in figure 10.1).

Astronomers have found many black holes with a size (that is, mass) of the order of our Sun (a bit larger, in fact). But there are also gigantic back holes. There is one of these at the centre of almost all of the galaxies, including our own.

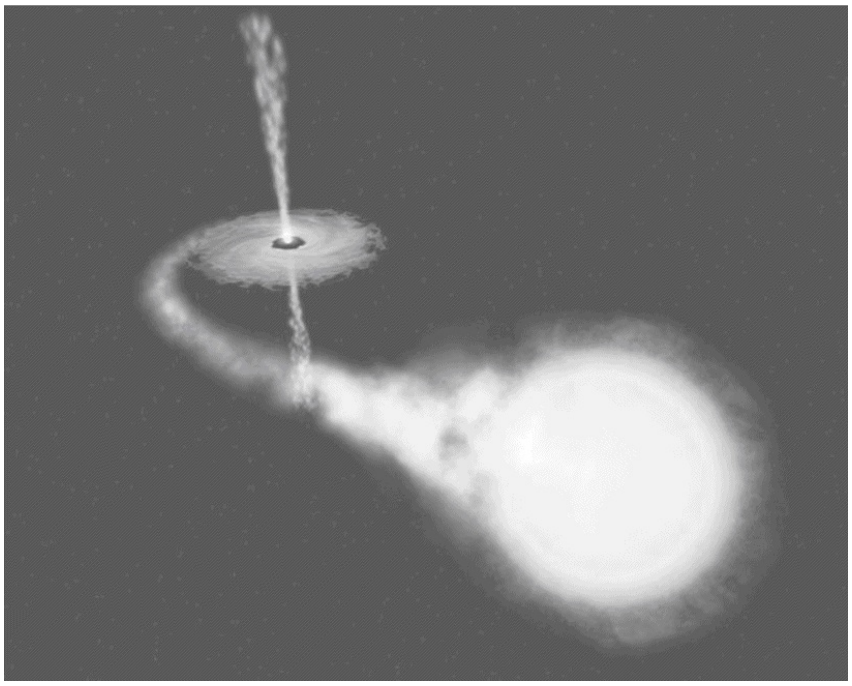


Figure 10.1 Representation of a couple star/black hole. The star loses matter, which is partly absorbed by the black hole, partly projected by it in jets in the direction of its poles.

The black hole at the centre of our own galaxy is currently being studied in detail. It has a mass a million times greater than our Sun. Every so often, a